User Churn Project | Two-Sample Hypothesis Test Results

Prepared for: Waze Leadership Team

Overview

Waze's data team is working on a project to build a machine learning model that predicts monthly user churn. As part of this project, leadership requested a statistical analysis to see if a user's phone type affects how much they use the app.

Objective

The main goal of this analysis was to determine if there is a statistically significant difference in the average (mean) number of monthly drives between iPhone users and Android users. This helps Waze decide if we need to build special app updates for a specific device type to keep users from leaving.

Results

- iPhone users in our sample averaged 67.86 drives per month, and Android users averaged 66.23 drives per month.
- A formal two-sample t-test resulted in a p-value of 0.1434. Because this is greater than 0.05, we fail to reject the null hypothesis.
- There is **no** statistically significant difference between the two groups. iPhone and Android users drive with Waze at basically the same rate. The tiny difference we saw is just random noise.

Next Steps

➔ **It is not recommended to waste time or money making unique product changes to just one version of the app to try and boost drive counts.**

➔ **When building the machine learning model to predict churn, treat `device` as a low-priority control feature.**

➔ **Pivot our modeling efforts toward user behavior features instead, such as how many days a user opens the app compared to how many kilometers they drive, as these provide a much stronger signal for churn.**